Supply Chain Specialization:

Quantifying North Carolina's Readiness for Clean Energy Investment — An Economic Base Analysis

To: Libby Lang, Director of Research; Elizabeth Sada, Contractor
From: Iain Holmes, Research Intern
Subject: Empirical Analysis of Southeastern Clean Energy Agglomeration (2-Year Avg: 2023–2024)

Multi-year empirical analysis of federal employment data reveals that North Carolina has established a dominant, rapidly accelerating cluster in clean energy and advanced manufacturing. Evaluated against a six-state Southeastern panel (North Carolina, South Carolina, Virginia, Tennessee, Georgia, and Florida — North Carolina's four bordering states plus Florida, its closest employment-size peer among Southeastern states) across 15 tracked industries, North Carolina demonstrates clear regional leadership within a critical "Core-5" subset of clean energy sectors. North Carolina's composite Location Quotient (LQ) for these Core-5 sectors stands at 1.65, well ahead of the next-closest peer, Florida, at 1.03. Composite LQ figures throughout this report are unweighted averages across the indicated sector group — each tracked NAICS code contributes equally regardless of its underlying employment level.

This multi-year employment baseline gives site-selection committees and energy-focused corporate executives verifiable evidence that North Carolina possesses a mature, stable, and dense vendor and contractor ecosystem, capable of deploying clean energy infrastructure more efficiently than competing Southeastern states.

> **CORE FINDING: The 'Core-5' Leadership Advantage**
> **Composite Dominance:** North Carolina leads the six-state panel with a Core-5 composite employment-based LQ of 1.65 — the highest of any state in the panel.
> **Outright Sector Leadership:** North Carolina ranks #1 across the panel in three of five Core-5 categories: Turbine & Power Transmission Equipment Mfg. (LQ: 2.28), Power & Communication Line Construction (LQ: 1.59), and Semiconductor & Related Device Mfg. (LQ: 0.95).
> **Panel-Leading Momentum:** North Carolina's Core-5 composite LQ rose from 1.21 in 2023 to 2.10 in 2024 — a +0.89 gain, the largest increase of any state in the panel over the same period.

Methodology & Data Framework

To evaluate regional specialization while avoiding federal employment data suppression, this analysis utilizes a multi-year pooled dataset from the U.S. Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW[1]) program (2023–2024 annual averages). Averaging across multiple years filters out single-year administrative reporting noise and temporary permitting spikes, confirming structural permanence of the workforce. This analysis is built primarily on **employment-based** location quotients, which measure the concentration of active workforce rather than the number of business establishments; a

[1]U.S. Bureau of Labor Statistics, Quarterly Census of Employment and Wages (QCEW) Program Overview. bls.gov/cew/

secondary **establishment-based** view is introduced later (see "Depth and Cost-Efficiency of the Cluster") to test whether that employment specialization is broad-based or concentrated among a few large employers.

- **The Metric:** The Employment Location Quotient compares the local concentration of employment in a specific industry against the national baseline.
- **The Baseline (LQ = 1.0):** Following standard regional economic definitions utilized by the U.S. Bureau of Economic Analysis (BEA[2]), an LQ of 1.0 indicates that a state's industrial concentration perfectly matches the national average.
- **The Cluster Threshold (LQ ≥ 1.25):** Grounded in standard targeted regional economic development (TRED[3]) frameworks, an LQ exceeding 1.25 denotes a statistically significant regional agglomeration ("super-cluster"): a thriving, highly concentrated network of skilled workers and specialized local suppliers. In this context, specialization means that incoming companies have immediate, local access to the exact contracting and technical expertise they need without relying on out-of-state resources; this dense ecosystem cultivates stronger businesses by driving down procurement costs through competitive bidding, minimizing project deployment delays, and fostering shared operational efficiencies.
- **Multi-Year Robust Baseline (2-Year Average LQ):** To filter out single-year administrative reporting noise, temporary permitting spikes, or localized cyclical volatility, this analysis constructs a robust multi-year baseline by averaging employment Location Quotients across consecutive annual datasets (2023 and 2024). This pooled approach confirms structural permanence rather than relying on a single annual snapshot.

ANALYTICAL STANDARDS & FRAMEWORK SUMMARY

Data Source: *U.S. BLS QCEW (2023–2024 annual averages, private ownership, own_code == 5), six-state panel, 15 clean-energy and advanced-manufacturing NAICS codes.*
Core Benchmarks: *LQ = 1.0 (national parity) | LQ ≥ 1.25 (significant regional cluster).*
Methodological Edge: *Multi-year pooling eliminates reporting anomalies and demonstrates structural, not transient, permanence.*

Regional specialization is quantified using the Employment Location Quotient, defined as the ratio of a state's industrial concentration to the national baseline:

$$\mathrm{LQ} = \frac{\mathrm{e_i, state}/\mathrm{E_{state}}}{\mathrm{e_i, US}/\mathrm{E_{US}}}$$

Where:

- $e_i, state$: Number of employees in target industry i within the state
- E_{state}: Total private-sector employment across all industries in the state
- e_i, US: Total national employment in target industry i
- E_{US}: Total national private-sector employment across all industries

[2]U.S. Bureau of Economic Analysis, "What are location quotients (LQs)?" bea.gov/help/faq/478
[3]Northeast Regional Center for Rural Development (NERCRD) & Penn State Extension, "Location Quotients and TRED." nercrd.psu.edu/location-quotients-and-tred/

SITE-SELECTION INTELLIGENCE: THE 1.25 CLUSTER IMPERATIVE

For corporate capital deployment, an LQ exceeding 1.25 serves as an empirical indication of supply-chain depth. Rather than import engineering talent or out-of-state contractors, incoming facilities plug directly into an active, competitive local vendor ecosystem[4] — translating directly into compressed construction timelines, lower subcontractor bids, and minimized execution risk.[5]

The figures and tables that follow visualize this LQ specialization using different key industry variables incorporated with the model. Among these: current market standing/framework, regional rank across all 15 clean energy sub-sectors, year-over-year momentum, wage premium and sub-sector depth and infrastructure. No single metric captures a state's readiness for clean energy investment. When the effects are compounded, however, patterns begin to emerge and a story develops. In the case of clean energy in the southeast, that story positions North Carolina as an incredibly competitive ecosystem in which to invest.

The existing market infrastructure, logistics networks, and historical progress due to government funding and sustained industry growth have all assisted in growing not only a multitude of adjacent markets within the clean energy space, but also a crucial amount of depth across these core-5 industries that the state is currently one of the most attractive locations within the region for business site selection.

North Carolina's Regional Position

Across the full 15-sector clean energy and advanced manufacturing basket, North Carolina's specialization is concentrated, without becoming stagnant. NC posts a composite LQ of 1.06 across all 15 tracked industries — third among the six-state panel behind Tennessee (1.46) and South Carolina (1.36) — while its Core-5 composite of 1.65 leads the panel outright. Figure 1 shows this pattern sector by sector across every tracked NAICS code; Figure 2 maps the same specialization geographically, for both the full 15-sector basket and the Core-5 leadership sectors alone.

[4]Marshall, A. (1890). *Principles of Economics.* Macmillan — the foundational treatment of localization economies: labor-market pooling, input-sharing, and knowledge spillovers among co-located firms.
[5]Porter, M.E. (1990). *The Competitive Advantage of Nations.* Free Press; companion article: "Clusters and the New Economics of Competition," *Harvard Business Review* (Nov–Dec 1998). hbr.org/1998/11/clusters-and-the-new-economics-of-competition

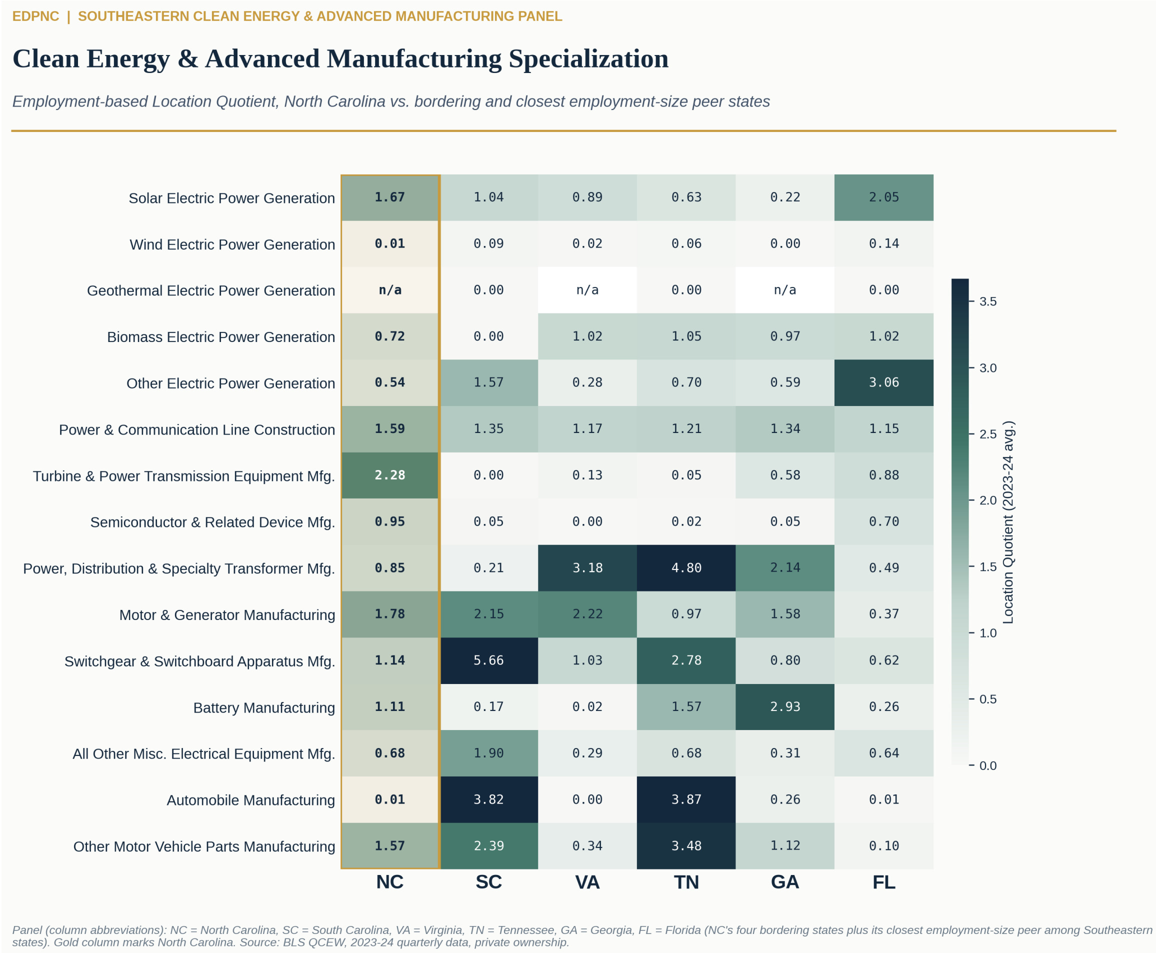

Figure 1: *Southeastern Clean Energy Specialization Heatmap. Employment-based LQ, all 15 tracked NAICS codes, North Carolina vs. the six-state panel.*

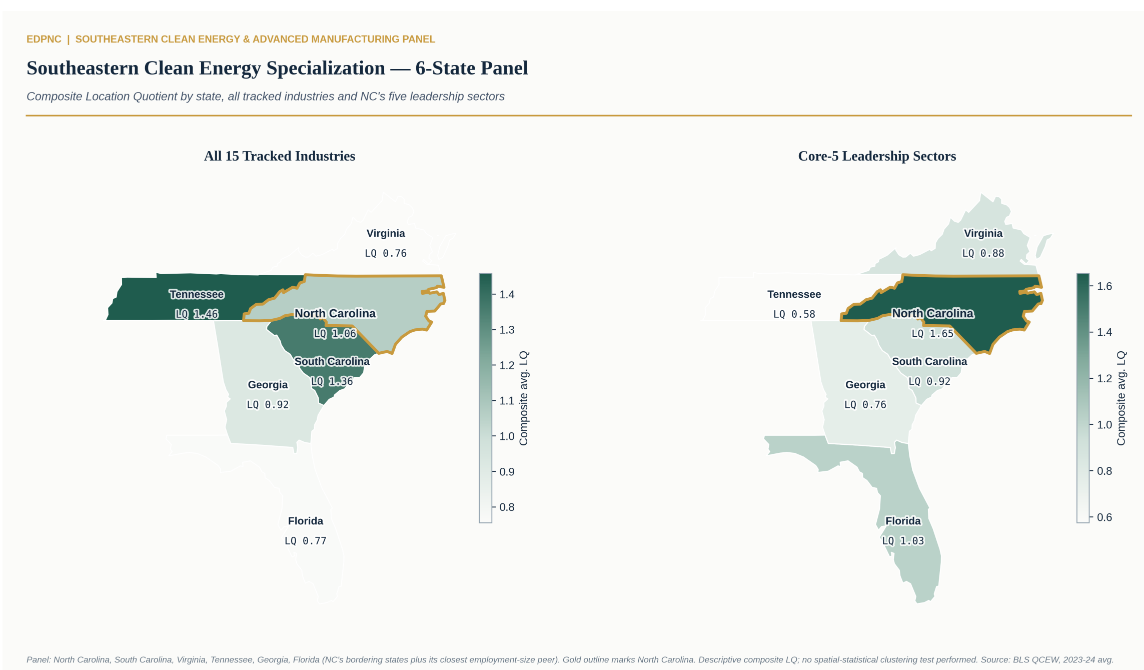

Figure 2: *Regional Specialization, Mapped. Composite LQ by state — all 15 industries (left) and the Core-5 leadership sectors alone (right).*

Sector-by-Sector Competitive Standing

Evaluating North Carolina alongside Florida, Georgia, South Carolina, Tennessee, and Virginia across the five Core-5 clean-energy supply-chain sub-sectors using combined 2023–2024 QCEW data yields distinct regional positioning:

- **Turbine & Power Transmission Equipment Mfg. (NAICS 333611) — 2-Yr Avg LQ: 2.28:** North Carolina leads the panel outright, clearing the 1.25 cluster threshold by a wide margin and outpacing its nearest competitor, Florida (LQ: 0.88), by 1.40 points.
- **Motor & Generator Manufacturing (NAICS 335312) — 2-Yr Avg LQ: 1.78:** North Carolina holds a strong specialization well above the cluster threshold, trailing only Virginia (2.22) and South Carolina (2.15) within the panel.
- **Solar Electric Power Generation (NAICS 221114) — 2-Yr Avg LQ: 1.67:** Operating well above the cluster threshold and ranking second in the panel behind Florida (2.05), this baseline ensures ample operational depth for incoming commercial solar generation projects.
- **Power & Communication Line Construction (NAICS 237130) — 2-Yr Avg LQ: 1.59:** North Carolina leads the panel outright in this construction and grid-interconnection category — the infrastructure most directly tied to solar and clean-energy deployment.
- **Semiconductor & Related Device Mfg. (NAICS 334413) — 2-Yr Avg LQ: 0.95:** A developing but panel-leading support sector that complements the primary deployment cluster; notably, this is the one Core-5 category where North Carolina leads the panel without yet clearing the 1.25 cluster threshold.
- **Panel-Leading Momentum:** North Carolina's Core-5 composite employment LQ rose from 1.21 in 2023 to 2.10 in 2024, a +0.89 gain — the largest increase of any state in the panel over the same period, while Florida, South Carolina, and Georgia each posted flat or declining Core-5 composites over the same window.

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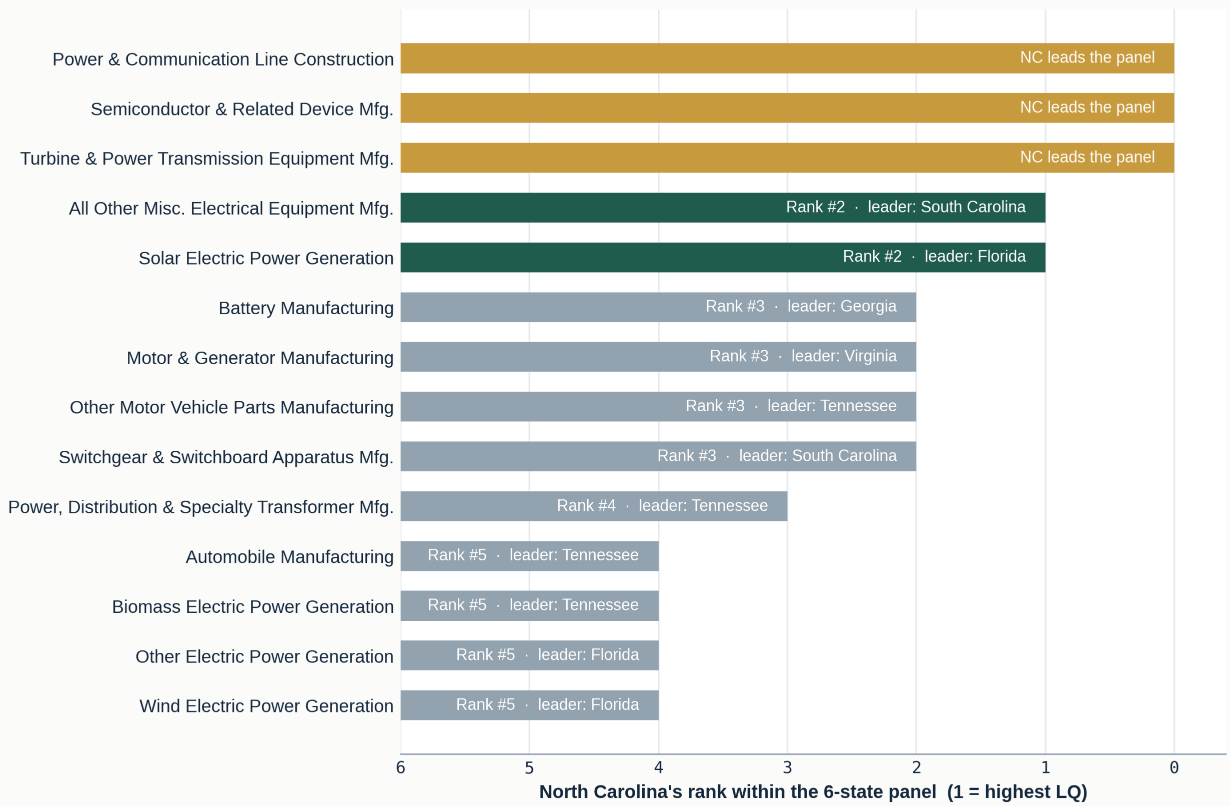
North Carolina's Competitive Position by Sector

Rank of NC's Location Quotient within the 6-state neighbor & peer panel, across all 15 tracked industries



Panel: North Carolina, South Carolina, Virginia, Tennessee, Georgia, Florida. Gold = NC leads the panel; green = NC ranks 2nd; slate = NC ranks 3rd-6th. Source: BLS QCEW, 2023-24 avg. LQ.

Figure 3: *North Carolina's Rank Within the Panel. North Carolina outright leads three of 15 tracked sectors and places second or third in several more.*

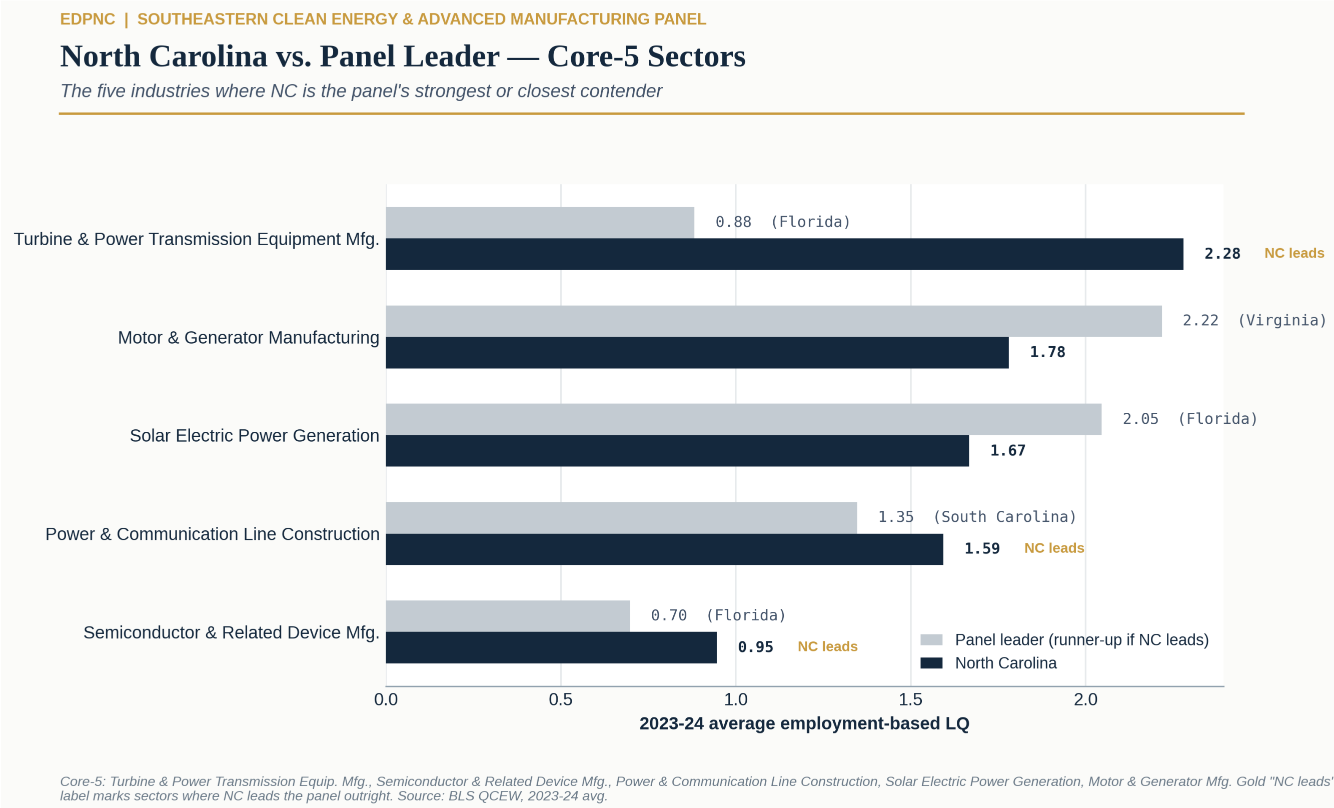


Figure 4: *North Carolina vs. Panel Leader, Core-5 Sectors. Where North Carolina leads, the margin over the nearest competitor is often decisive — 1.40 points in Turbine & Power Transmission Equipment Mfg. alone.*

Momentum: Where Specialization Is Accelerating

Current positioning tells only part of the story — direction of travel matters just as much for a multi-year site-selection decision. Figure 5 tracks each panel state's composite LQ from 2023 to 2024: North Carolina's line rises more steeply than any peer's, both across the full 15-sector basket (+0.28) and, more dramatically, within the Core-5 leadership sectors (+0.89). Figure 6 maps this same momentum geographically, sizing each state's marker to the magnitude of its 2023–2024 change.

This divergence is consistent with a relative shift in specialized clean-energy employment toward North Carolina, relative to its regional peers; it reflects a change in each state's *share* of national industry employment and does not, by itself, establish that workers physically relocated — confirming that would require separate interstate migration-flow data.

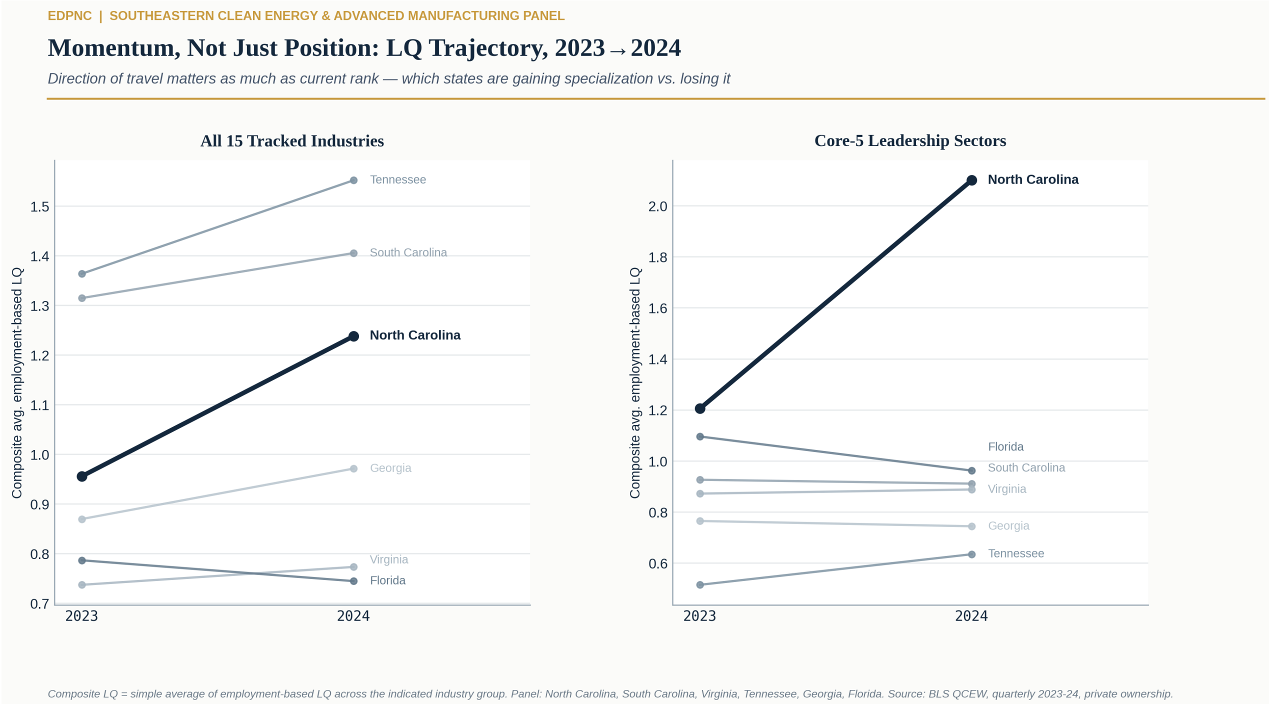


Figure 5: *Year-over-Year Composite LQ Trajectory, 2023→2024. North Carolina's gain outpaces every peer state in both the all-15 and Core-5 baskets.*

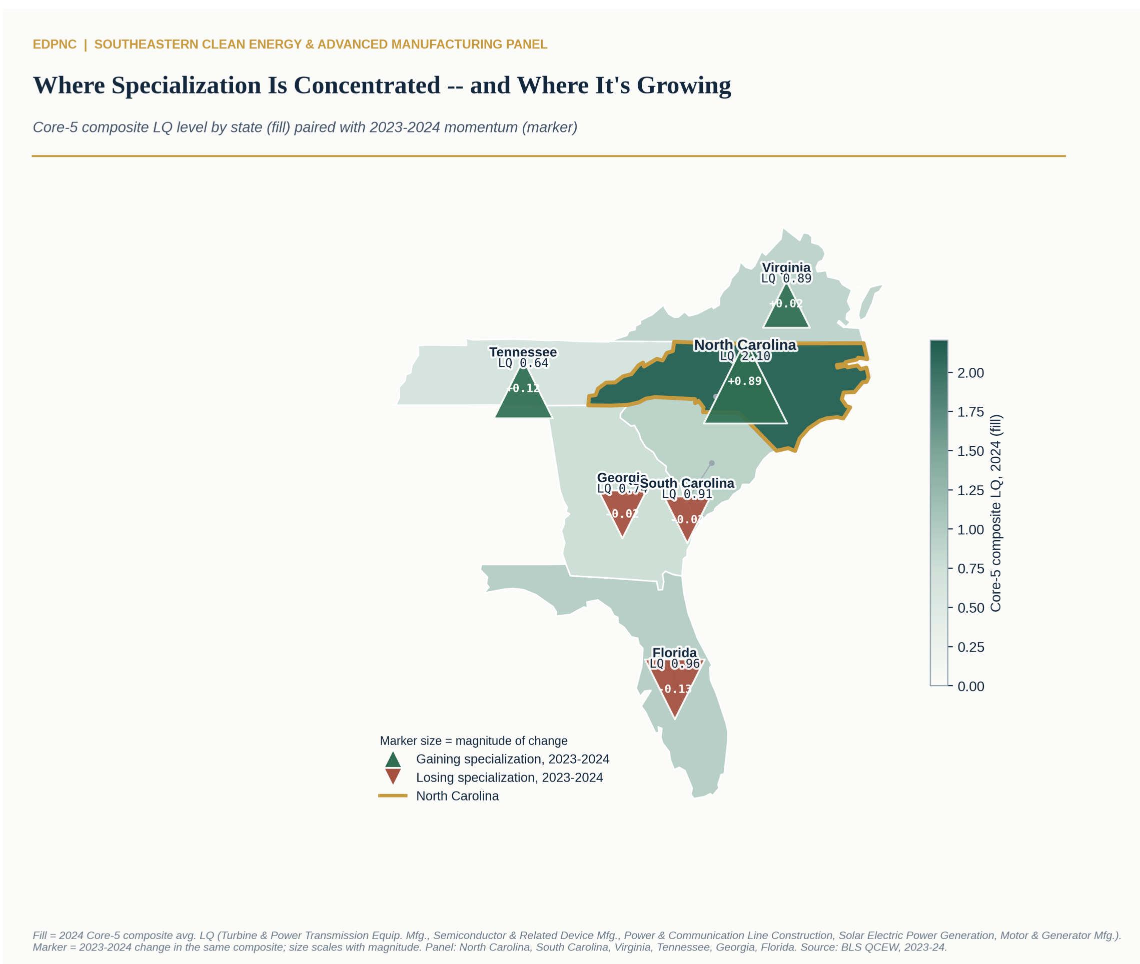


Figure 6: *Specialization Level and Momentum, Mapped. State fill shows 2024 Core-5 composite LQ; marker shows the 2023–2024 change, sized to magnitude.*

Depth and Cost-Efficiency of the Cluster

Two further questions matter to a site-selection committee beyond raw specialization: is this workforce affordable relative to peer states, and is it distributed across a resilient base of many vendors rather than concentrated in a handful of large, single-point-of-failure employers? Figure 7 positions each Core-5 sector's LQ against its average annual wage. North Carolina frequently combines a high LQ with a wage level below several of its peers — in Turbine & Power Transmission Equipment Mfg., for example, North Carolina posts the panel's highest LQ (2.28) alongside its **lowest** average annual wage of the six states. This is consistent with the classical view that regional wage differences largely reflect cost-of-living and amenity tradeoffs rather than specialization itself commanding a wage premium[6] — a meaningful cost-efficiency signal for incoming employers.

Figure 8 turns to establishment-based LQ and average establishment size. North Carolina's specialization in Power & Communication Line Construction, for instance, is built on a broad base of roughly **625 reporting establishments** averaging **19 employees** each, rather than a small number of mega-sites — a distributed structure associated in the regional economics literature with more resilient, competitively priced local supply chains[7]. By contrast, North Carolina's Turbine & Power Transmission and Semiconductor specializations are built on fewer, larger establishments (averaging 107 and 117 employees respectively) — a different but equally legitimate structure, more typical of capital-intensive advanced manufacturing than of a construction trade.

[6] Roback, J. (1982). "Wages, Rents, and the Quality of Life." *Journal of Political Economy, 90*(6), 1257–1278; see also Moretti, E. (2012). *The New Geography of Jobs.* Houghton Mifflin Harcourt — spatial-equilibrium theory explaining why regional wage differences reflect cost-of-living and amenity tradeoffs rather than specialization itself commanding a premium.

[7] Glaeser, E., Kallal, H., Scheinkman, J., & Shleifer, A. (1992). "Growth in Cities." *Journal of Political Economy, 100*(6), 1126–1152 — empirical test of whether industry growth is better explained by concentration among many small establishments (Marshall-Arrow-Romer externalities) or by cross-industry diversity (Jacobs externalities).

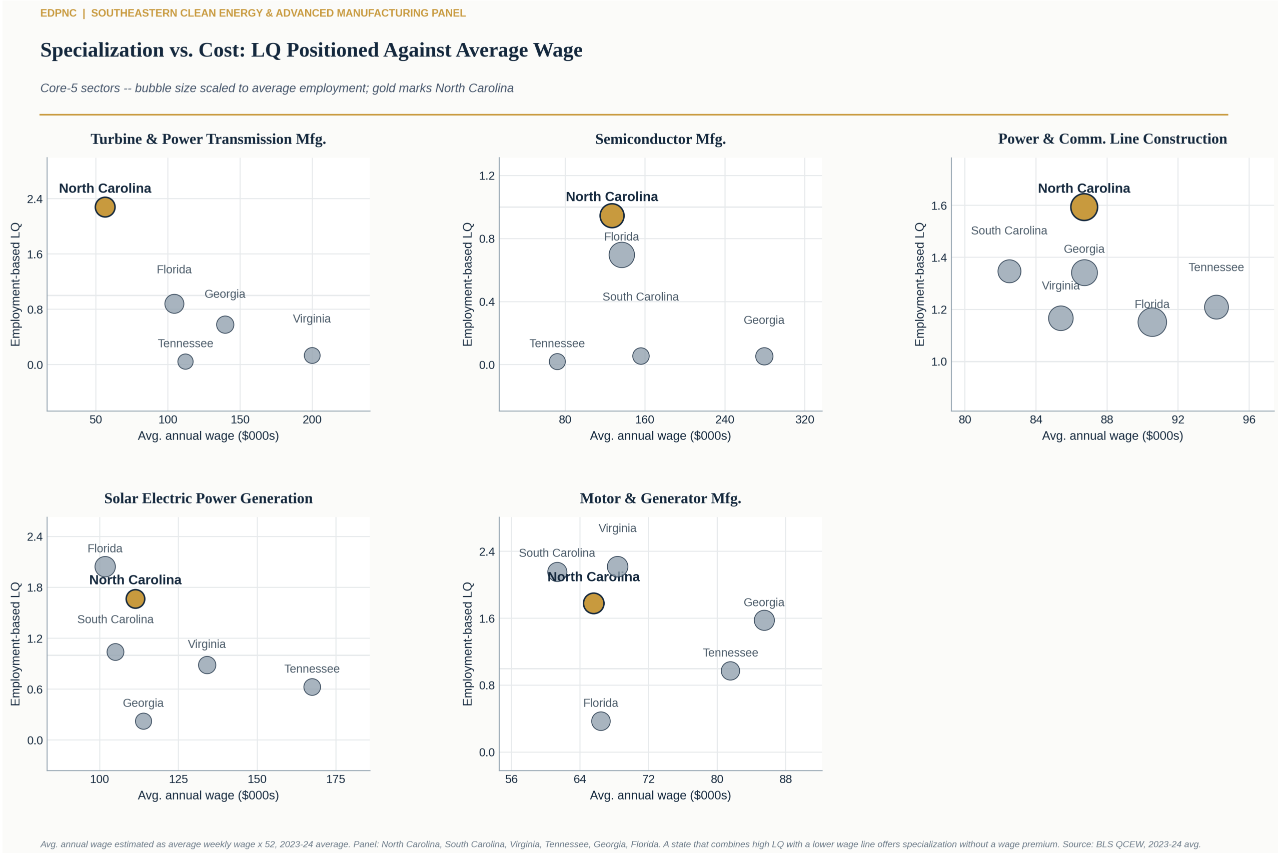


Figure 7: *Specialization vs. Cost. LQ positioned against average annual wage, Core-5 sectors; bubble size scaled to average employment. North Carolina frequently combines a high LQ with a competitive wage level.*

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Beneath the Employment Numbers: Establishment Density & Scale

Is NC's specialization built on many establishments or a few large employers, compared to peers?

Establishment-Based Location Quotient

	NC	SC	VA	TN	GA	FL
Solar Electric Power Generation	**1.00**	2.52	1.75	1.00	0.51	1.55
Wind Electric Power Generation	**0.17**	0.82	0.58	0.73	0.05	0.38
Geothermal Electric Power Generation	**n/a**	1.04	n/a	1.09	n/a	0.86
Biomass Electric Power Generation	**1.33**	1.18	2.96	1.10	0.83	0.71
Other Electric Power Generation	**0.54**	2.69	1.43	1.09	0.71	1.11
Power & Communication Line Construction	**1.54**	1.13	1.11	1.34	1.14	0.85
Turbine & Power Transmission Equipment Mfg.	**1.01**	3.58	1.09	1.49	1.38	0.93
Semiconductor & Related Device Mfg.	**0.68**	1.07	0.80	0.58	0.50	0.67
Power, Distribution & Specialty Transformer Mfg.	**0.70**	2.21	1.83	2.03	2.27	1.19
Motor & Generator Manufacturing	**1.10**	1.60	1.19	1.78	1.00	0.93
Switchgear & Switchboard Apparatus Mfg.	**0.88**	2.69	0.78	1.92	1.21	0.60
Battery Manufacturing	**0.97**	1.48	0.61	2.42	1.70	0.92
All Other Misc. Electrical Equipment Mfg.	**0.53**	3.12	1.29	0.97	1.08	1.29
Automobile Manufacturing	**0.89**	3.21	0.63	3.64	1.15	0.81
Other Motor Vehicle Parts Manufacturing	**0.75**	2.20	0.56	2.35	0.68	0.57

Establishment-based LQ

Average Establishment Size

	NC	SC	VA	TN	GA	FL
Solar Electric Power Generation	**11.3**	2.7	3.4	4.8	2.9	7.7
Wind Electric Power Generation	**0.8**	1.6	0.4	1.4	0.0	4.7
Geothermal Electric Power Generation	**n/a**	0.0	n/a	0.0	n/a	0.0
Biomass Electric Power Generation	**6.0**	0.0	3.8	11.9	12.5	13.7
Other Electric Power Generation	**6.6**	3.6	1.3	4.5	5.0	14.8
Power & Communication Line Construction	**19.3**	21.1	19.3	18.9	21.1	21.7
Turbine & Power Transmission Equipment Mfg.	**107.1**	0.0	5.5	1.6	18.0	36.8
Semiconductor & Related Device Mfg.	**117.2**	4.1	0.0	3.5	8.8	74.4
Power, Distribution & Specialty Transformer Mfg.	**63.7**	4.6	89.8	138.6	47.2	18.5
Motor & Generator Manufacturing	**75.1**	59.3	86.0	28.5	70.3	15.9
Switchgear & Switchboard Apparatus Mfg.	**67.3**	105.0	69.1	86.4	33.2	46.5
Battery Manufacturing	**77.1**	7.5	1.6	49.1	108.4	15.9
All Other Misc. Electrical Equipment Mfg.	**22.5**	10.0	3.8	13.5	4.7	7.4
Automobile Manufacturing	**5.1**	482.1	0.0	507.9	93.0	5.9
Other Motor Vehicle Parts Manufacturing	**158.8**	78.1	44.9	125.3	118.8	11.5

Avg. employees / establishment

Panel (column abbreviations): NC = North Carolina, SC = South Carolina, VA = Virginia, TN = Tennessee, GA = Georgia, FL = Florida. Establishment-based LQ uses count of reporting establishments in place of employment as the LQ input; avg. establishment size = avg. employment / avg. establishment count. Gold column marks North Carolina. Source: BLS QCEW, 2023-24 avg.

Figure 8: *Establishment Density & Scale. Establishment-based LQ (left) and average establishment size (right) reveal whether specialization rests on a broad vendor base or a few large employers.*

Data Sources & Access

This analysis relies on official public datasets published by federal statistical agencies to evaluate industrial employment concentration across Southeastern states.

1. U.S. Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW) — 2023 & 2024 Annual Data: The primary dataset utilized to query private-ownership employment (own_code == 5) across 15 target 6-digit NAICS codes spanning clean-energy generation, equipment manufacturing, and grid-construction, for North Carolina, South Carolina, Georgia, Virginia, Tennessee, Florida, and the U.S. baseline.

- Online Access: BLS QCEW Open Data Files

References

1. **U.S. Bureau of Economic Analysis (BEA) — Location Quotient Methodology:** Official federal guidance defining Location Quotients as analytical ratios used to compare regional industrial concentration against a national reference baseline.
   - Online Access: BEA Location Quotient FAQ
2. **U.S. Bureau of Labor Statistics (BLS) — QCEW Program Overview:** Comprehensive program outlining the collection, structure, and economic applications of quarterly establishment and employment data.
   - Online Access: BLS QCEW Home Page
3. **Northeast Regional Center for Rural Development (NERCRD) & Penn State Extension — Targeted Regional Economic Development (TRED):** Regional economic framework detailing how Location Quotients and threshold benchmarks (such as LQ ≥ 1.25) are applied to identify export-oriented industrial clusters.
   - Online Access: NERCRD Location Quotients and TRED
4. **Marshall, A. (1890). Principles of Economics:** The foundational academic source for agglomeration (localization) economies — labor pooling, input-sharing, and knowledge spillovers among co-located firms — underlying the cluster logic used throughout this analysis.
5. **Porter, M.E. (1990). The Competitive Advantage of Nations:** Foundational framework underlying the site-selection logic that dense, geographically concentrated supplier and labor networks reduce cost and risk for firms locating within a cluster.
   - Online Access: HBR — Clusters and the New Economics of Competition
6. **Glaeser, Kallal, Scheinkman & Shleifer (1992). “Growth in Cities,” Journal of Political Economy:** Empirical test distinguishing localization externalities (many establishments in one industry) from diversity externalities, underlying the establishment-density analysis in this report.
7. **Roback, J. (1982), “Wages, Rents, and the Quality of Life”; Moretti, E. (2012), The New Geography of Jobs:** Spatial-equilibrium theory explaining why regional wage differences reflect cost-of-living and amenity tradeoffs rather than specialization commanding a premium.
8. **EDPNC — Commercial Property Assessed Capital Expenditure (C-PACE) Program:** Program overview for the financing mechanism referenced under Strategic Implications, allowing commercial property owners to finance energy efficiency, renewable energy, and resiliency improvements through long-term, low-interest assessments.
   - Online Access: EDPNC C-PACE Program